Assignment no 18

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Domain : AI & ML

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Q1.

```
First 10 Rows:
     model  year  price transmission  mileage fuelType  tax   mpg  engineSize
0   Fiesta  2017  12000    Automatic    15944   Petrol  150  57.7         1.0
1    Focus  2018  14000       Manual     9083   Petrol  150  57.7         1.0
2    Focus  2017  13000       Manual    12456   Petrol  150  57.7         1.0
3   Fiesta  2019  17500       Manual    10460   Petrol  145  40.3         1.5
4   Fiesta  2019  16500    Automatic     1482   Petrol  145  48.7         1.0
5   Fiesta  2015  10500       Manual    35432   Petrol  145  47.9         1.6
6     Puma  2019  22500       Manual     2029   Petrol  145  50.4         1.0
7   Fiesta  2017   9000       Manual    13054   Petrol  145  54.3         1.2
8     Kuga  2019  25500    Automatic     6894   Diesel  145  42.2         2.0
9    Focus  2018  10000       Manual    48141   Petrol  145  61.4         1.0

Last 5 Rows:
        model  year  price transmission  mileage fuelType  tax   mpg  \
17961   B-MAX  2017   8999       Manual    16700   Petrol  150  47.1
17962   B-MAX  2014   7499       Manual    40700   Petrol   30  57.7
17963   Focus  2015   9999       Manual     7010   Diesel   20  67.3
17964      KA  2018   8299       Manual     5007   Petrol  145  57.7
17965   Focus  2015   8299       Manual     5007   Petrol   22  57.7

       engineSize
17961         1.4
17962         1.0
17963         1.6
17964         1.2
17965         1.0

Shape of Dataset:
(17966, 9)
```

```
Number of Rows: 17966
Number of Columns: 9

Data Types of Columns:
model            object
year              int64
price             int64
transmission     object
mileage           int64
fuelType         object
tax               int64
mpg             float64
engineSize      float64
dtype: object
```

Q2.

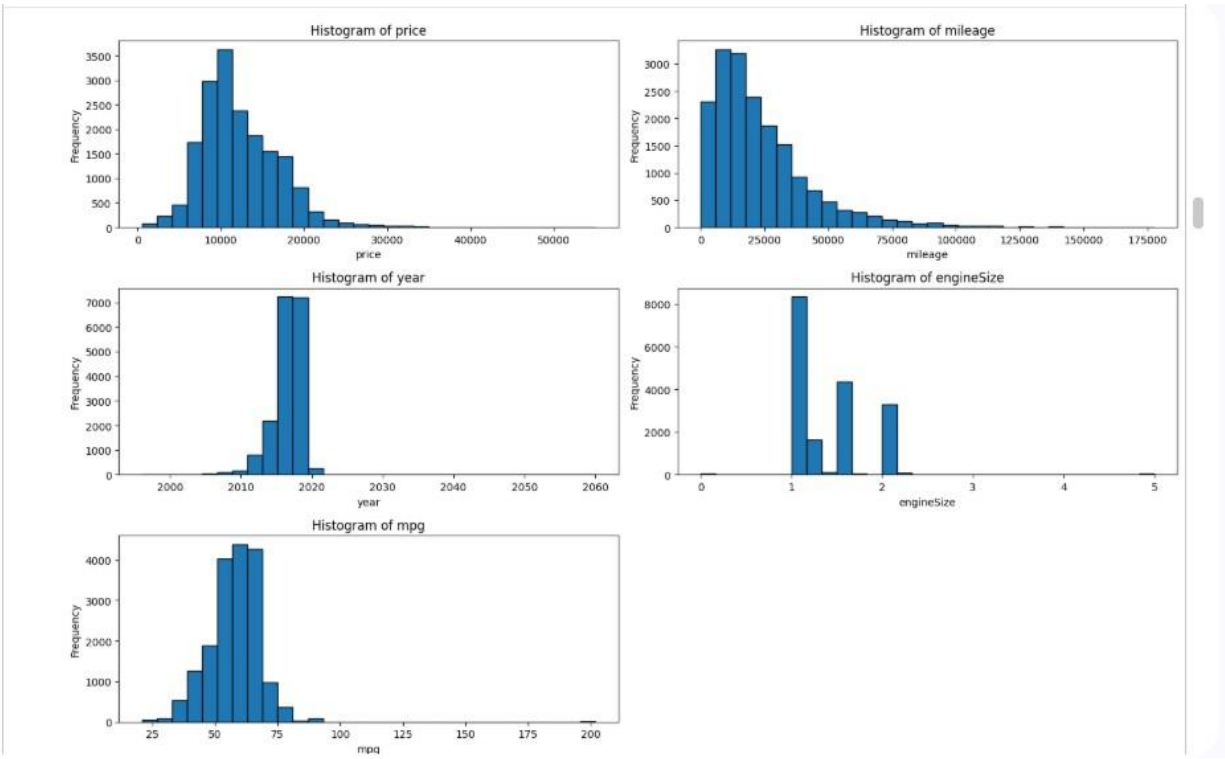
```
Missing Values in Each Column:
model           0
year            0
price           0
transmission    0
mileage         0
fuelType        0
tax             0
mpg             0
engineSize      0
dtype: int64

Total Missing Values: 0

Number of Duplicate Rows: 154

Dataset Shape After Removing Duplicates: (17812, 9)
```

Q3.

```
Statistical Summary of Numeric Columns:
...
                year         price        mileage           tax           mpg  \
count  17966.000000  17966.000000   17966.000000  17966.000000  17966.000000
mean    2016.866470  12279.534844   23362.608761    113.329456     57.906980
std        2.050336   4741.343657   19472.054349     62.012456     10.125696
min     1996.000000    495.000000       1.000000      0.000000     20.800000
25%     2016.000000   8999.000000    9987.000000     30.000000     52.300000
50%     2017.000000  11291.000000   18242.500000    145.000000     58.900000
75%     2018.000000  15299.000000   31060.000000    145.000000     65.700000
max     2060.000000  54995.000000  177644.000000    580.000000    201.800000

         engineSize
count  17966.000000
mean       1.350807
std        0.432367
min        0.000000
25%        1.000000
50%        1.200000
75%        1.500000
max        5.000000

Minimum, Maximum, Mean, and Median:

PRICE
Minimum : 495
Maximum : 54995
Mean    : 12279.534843593454
Median  : 11291.0
------------------------------
```

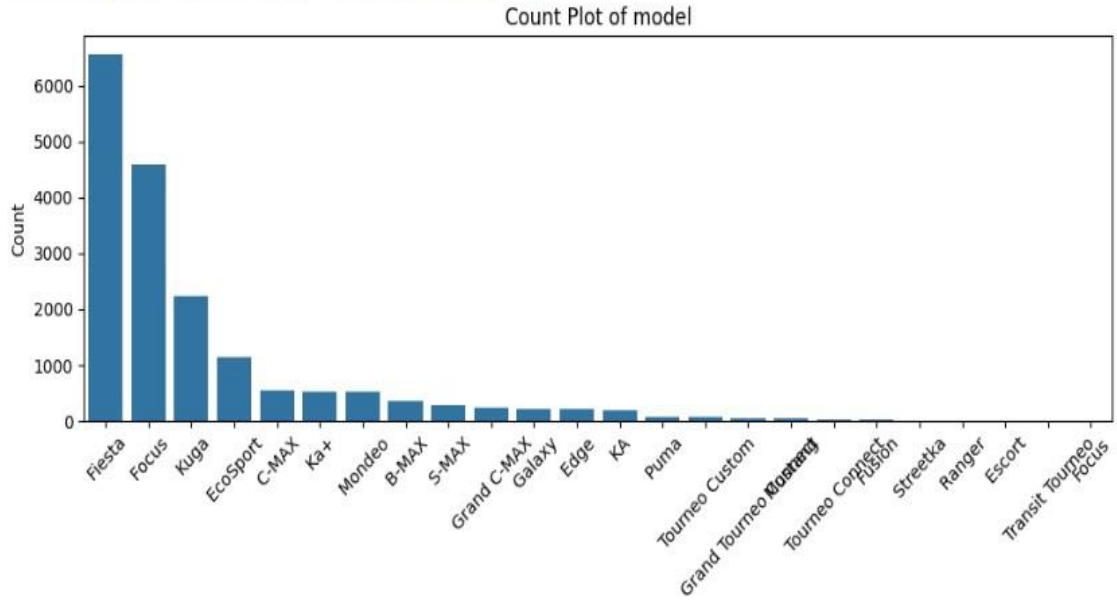
```
MILEAGE
Minimum : 1
Maximum : 177644
Mean    : 23362.608760992985
Median  : 18242.5
------------------------------
YEAR
Minimum : 1996
Maximum : 2060
Mean    : 2016.8664699988867
Median  : 2017.0
------------------------------
```

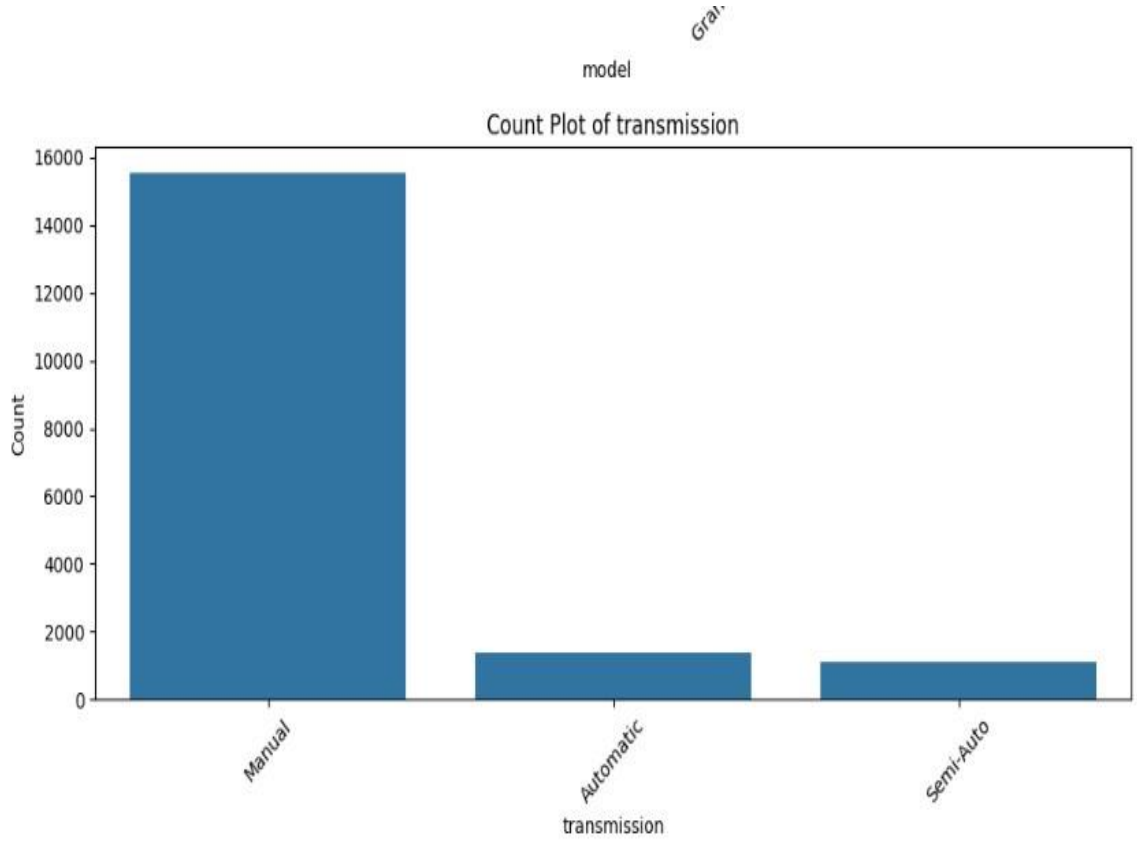
Q4

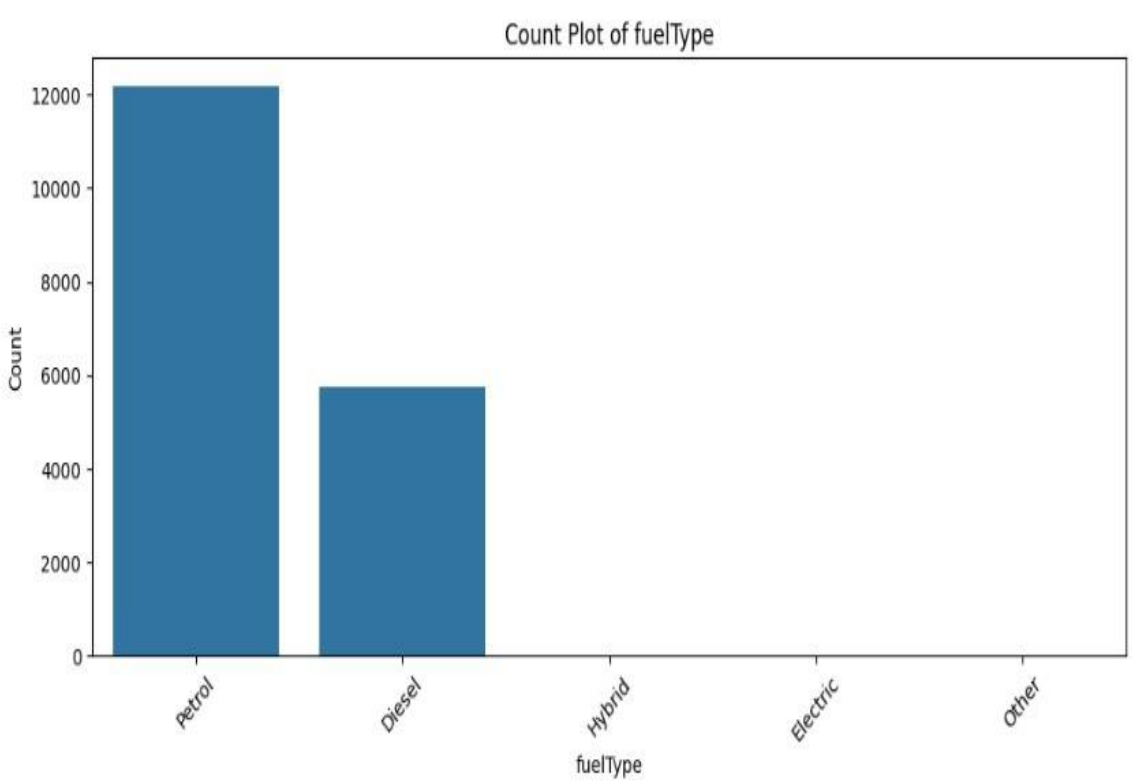

Histogram of price
Frequency
price
Histogram of mileage
Frequency
mileage
Histogram of year
Frequency
year
Histogram of engineSize
Frequency
engineSize
Histogram of mpg
Frequency
mpg

Q5

```
Categorical Columns:
Index(['model', 'transmission', 'fuelType'], dtype='object')
```




model
Count Plot of transmission
Count
16000
14000
12000
10000
8000
6000
4000
2000
0
Manual
Automatic
Semi-Auto
transmission

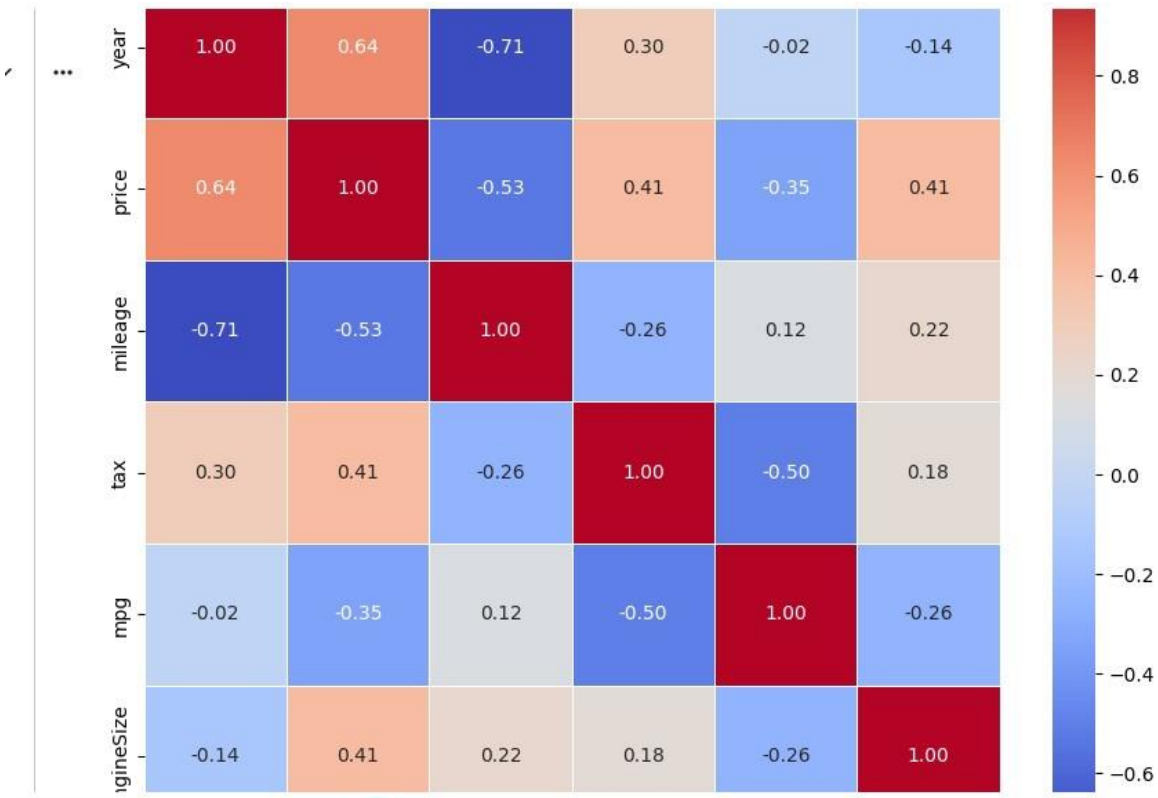

Count Plot of fuelType
12000
10000
8000
6000
4000
2000
0
Count
Petrol
Diesel
Hybrid
Electric
Other
fuelType

```
Value Counts for model:
model
 Fiesta                  6557
 Focus                   4588
 Kuga                    2225
 EcoSport                1143
 C-MAX                    543
 Ka+                      531
 Mondeo                   526
 B-MAX                    355
 S-MAX                    296
 Grand C-MAX              247
 Galaxy                   228
 Edge                     208
 KA                       199
 Puma                      80
 Tourneo Custom            69
 Grand Tourneo Connect     59
 Mustang                   57
 Tourneo Connect           33
 Fusion                    16
 Streetka                   2
 Ranger                     1
 Escort                     1
 Transit Tourneo            1
Focus                       1
Name: count, dtype: int64
```

```
Value Counts for transmission:
transmission
Manual       15518
Automatic     1361
Semi-Auto     1087
Name: count, dtype: int64

Value Counts for fuelType:
fuelType
Petrol      12179
Diesel       5762
Hybrid         22
Electric        2
Other           1
Name: count, dtype: int64
```

Q6.

```
Correlation Matrix:
               year     price   mileage       tax       mpg  engineSize
year       1.000000  0.636009 -0.707818  0.298506 -0.022967   -0.137311
price      0.636009  1.000000 -0.530659  0.406857 -0.346419    0.411178
mileage   -0.707818 -0.530659  1.000000 -0.260451  0.120075    0.215035
tax        0.298506  0.406857 -0.260451  1.000000 -0.502980    0.184363
mpg       -0.022967 -0.346419  0.120075 -0.502980  1.000000   -0.260521
engineSize -0.137311  0.411178  0.215035  0.184363 -0.260521    1.000000
```


year
price
mileage
tax
mpg
gineSize
1.00
0.64
-0.71
0.30
-0.02
-0.14
0.64
1.00
-0.53
0.41
-0.35
0.41
-0.71
-0.53
1.00
-0.26
0.12
0.22
0.30
0.41
-0.26
1.00
-0.50
0.18
-0.02
-0.35
0.12
-0.50
1.00
-0.26
-0.14
0.41
0.22
0.18
-0.26
1.00
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6

```
Correlation with Price:
price         1.000000
year          0.636009
engineSize    0.411178
tax           0.406857
mpg          -0.346419
mileage      -0.530659
Name: price, dtype: float64
```

Q7.

```
Dependent (Target) Feature:
price

Independent (Input) Features:
model
year
transmission
mileage
fuelType
tax
mpg
engineSize
```

Q8.

```
Categorical Columns:
Index(['model', 'transmission', 'fuelType'], dtype='object')

Before One-Hot Encoding (First 5 Rows):
  fuelType transmission
0   Petrol    Automatic
1   Petrol       Manual
2   Petrol       Manual
3   Petrol       Manual
4   Petrol    Automatic

After One-Hot Encoding (First 5 Rows):
   transmission_Manual  transmission_Semi-Auto  fuelType_Electric  \
0                False                   False              False
1                 True                   False              False
2                 True                   False              False
3                 True                   False              False
4                False                   False              False

   fuelType_Hybrid  fuelType_Other  fuelType_Petrol
0            False           False             True
1            False           False             True
2            False           False             True
3            False           False             True
4            False           False             True
```

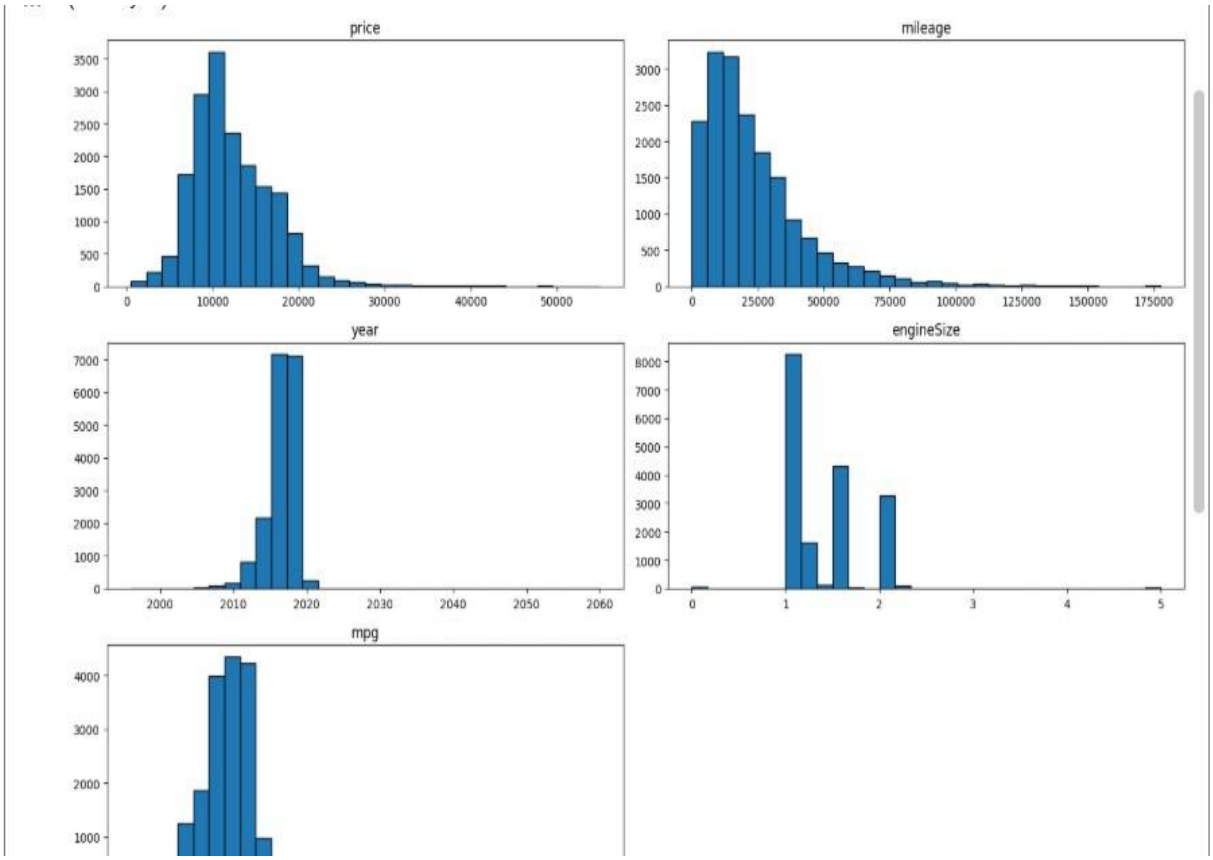
```
Original Dataset Shape : (17966, 9)
Encoded Dataset Shape  : (17966, 35)

Encoded Dataset Columns:
Index(['year', 'price', 'mileage', 'tax', 'mpg', 'engineSize', 'model_ C-MAX',
       'model_ EcoSport', 'model_ Edge', 'model_ Escort', 'model_ Fiesta',
       'model_ Focus', 'model_ Fusion', 'model_ Galaxy', 'model_ Grand C-MAX',
       'model_ Grand Tourneo Connect', 'model_ KA', 'model_ Ka+',
       'model_ Kuga', 'model_ Mondeo', 'model_ Mustang', 'model_ Puma',
       'model_ Ranger', 'model_ S-MAX', 'model_ Streetka',
       'model_ Tourneo Connect', 'model_ Tourneo Custom',
       'model_ Transit Tourneo', 'model_Focus', 'transmission_Manual',
       'transmission_Semi-Auto', 'fuelType_Electric', 'fuelType_Hybrid',
       'fuelType_Other', 'fuelType_Petrol'],
      dtype='object')
```

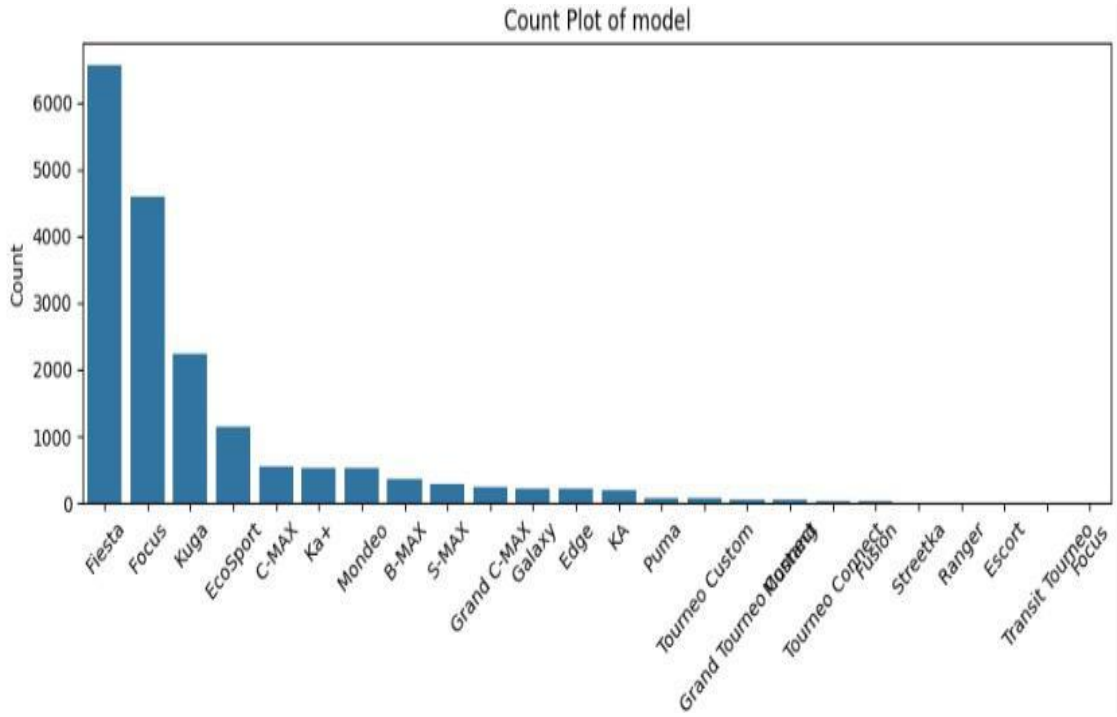
Q9.

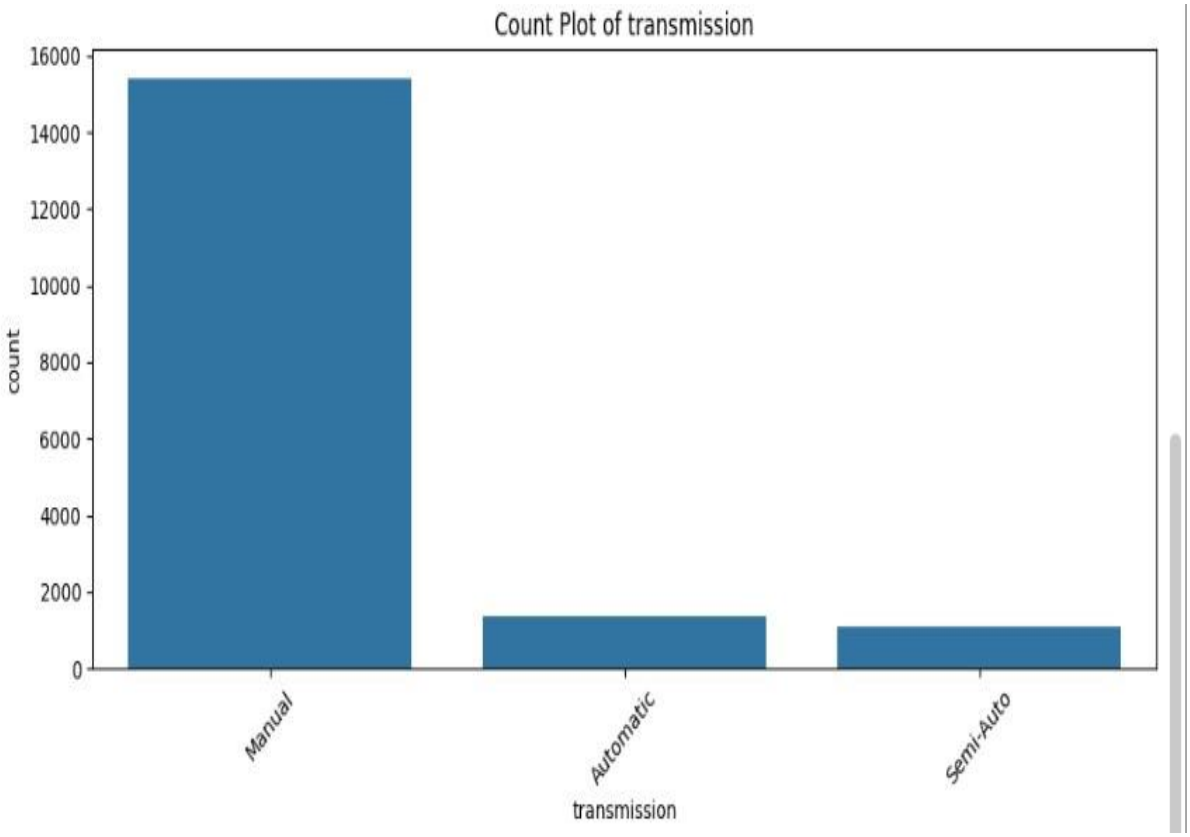
```
First 5 Rows of Scaled Data:
       year   mileage       tax       mpg  engineSize  model_ C-MAX  \
0  0.065128 -0.380998  0.591358 -0.020442   -0.811386         False
1  0.552866 -0.733359  0.591358 -0.020442   -0.811386         False
2  0.065128 -0.560132  0.591358 -0.020442   -0.811386         False
3  1.040605 -0.662640  0.510727 -1.738890    0.345070         False
4  1.040605 -1.123724  0.510727 -0.909294   -0.811386         False

   model_ EcoSport  model_ Edge  model_ Escort  model_ Fiesta  ...  \
0           False        False          False           True  ...
1           False        False          False          False  ...
2           False        False          False          False  ...
3           False        False          False           True  ...
4           False        False          False           True  ...

   model_ Tourneo Connect  model_ Tourneo Custom  model_ Transit Tourneo  \
0                   False                  False                   False
1                   False                  False                   False
2                   False                  False                   False
3                   False                  False                   False
4                   False                  False                   False

   model_Focus  transmission_Manual  transmission_Semi-Auto  \
0        False                False                   False
1        False                 True                   False
2        False                 True                   False
3        False                 True                   False
4        False                False                   False

   fuelType_Electric  fuelType_Hybrid  fuelType_Other  fuelType_Petrol
0              False            False           False             True
1              False            False           False             True
```

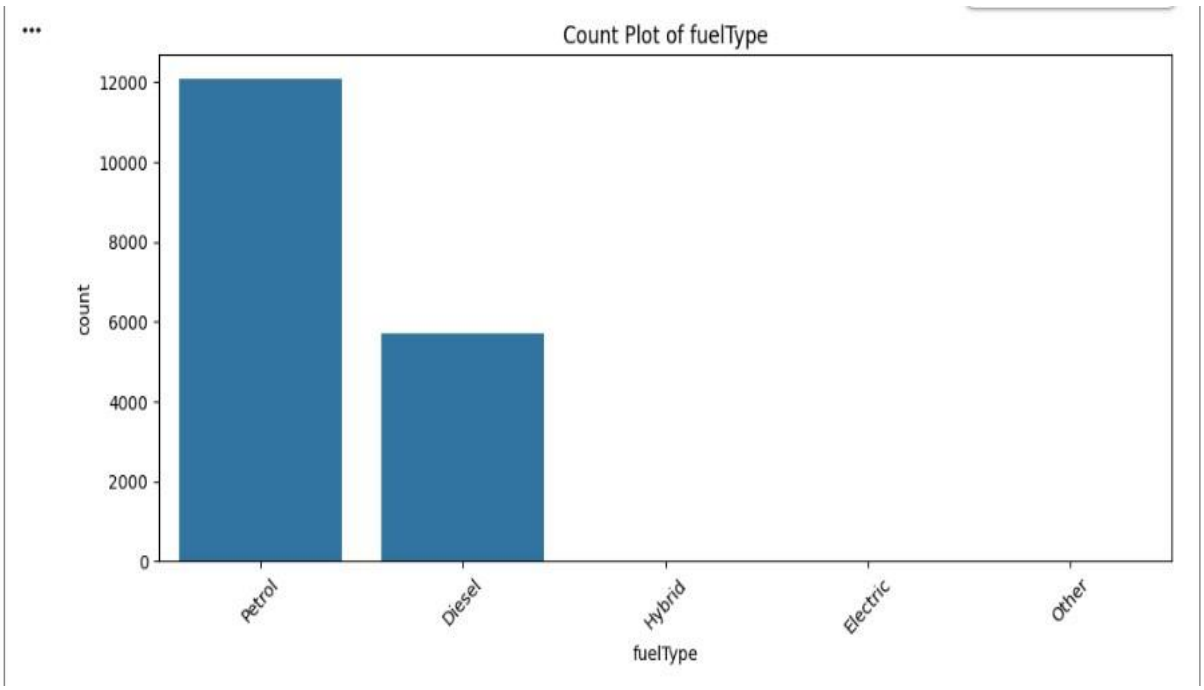
Q10.

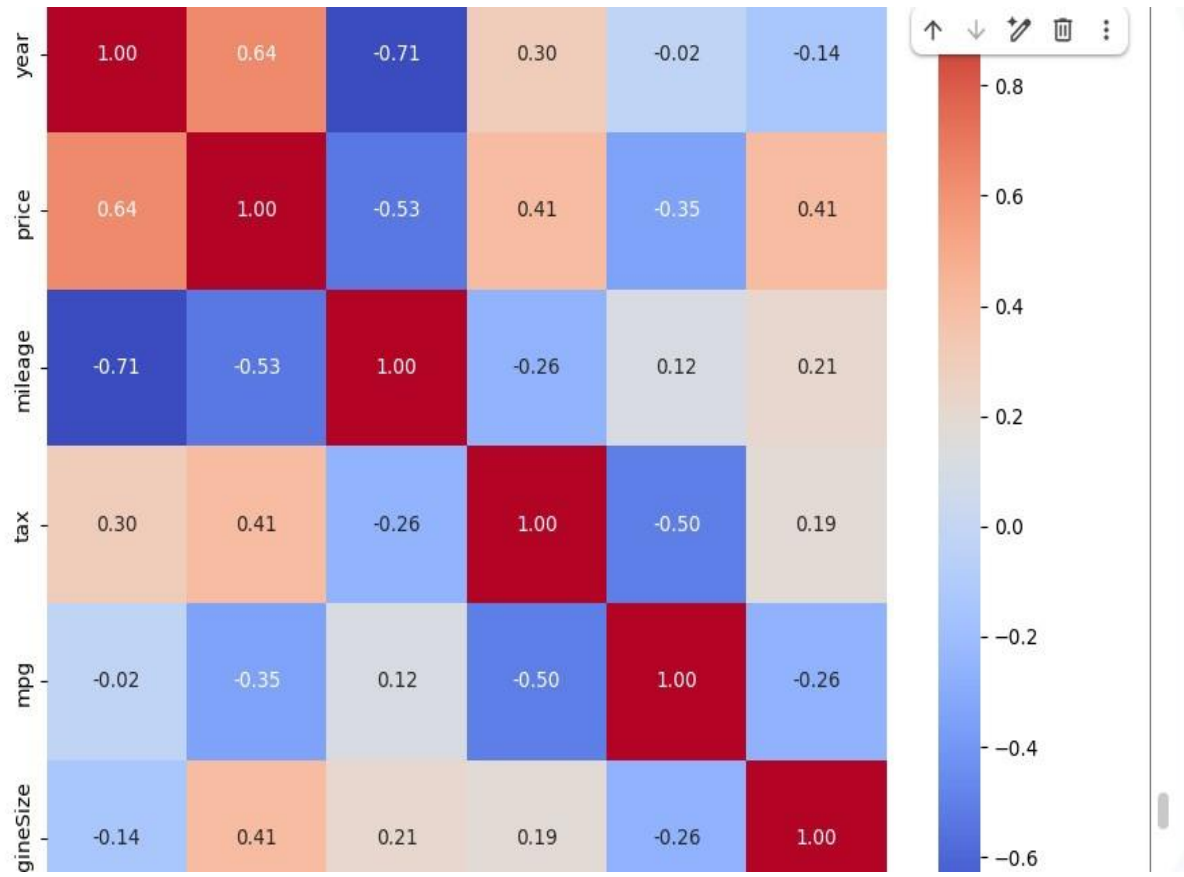
```
First 5 Rows:
     model  year  price transmission  mileage fuelType  tax   mpg  engineSize
0   Fiesta  2017  12000    Automatic    15944   Petrol  150  57.7         1.0
1    Focus  2018  14000       Manual     9083   Petrol  150  57.7         1.0
2    Focus  2017  13000       Manual    12456   Petrol  150  57.7         1.0
3   Fiesta  2019  17500       Manual    10460   Petrol  145  40.3         1.5
4   Fiesta  2019  16500    Automatic     1482   Petrol  145  48.7         1.0

Dataset Shape: (17966, 9)

Missing Values:
model           0
year            0
price           0
transmission    0
mileage         0
fuelType        0
tax             0
mpg             0
engineSize      0
dtype: int64

Duplicate Rows: 154

Shape After Removing Duplicates:
(17812, 9)
```


price
mileage
year
engineSize
mpg


Count Plot of model
Count
6000
5000
4000
3000
2000
1000
0
Fiesta
Focus
Kuga
EcoSport
C-MAX
Ka+
Mondeo
B-MAX
S-MAX
Grand C-MAX
Galaxy
Edge
KA
Puma
Tourneo Custom
Grand Tourneo
Tourneo Connect
Fusion
Streetka
Ranger
Escort
Transit Tourneo
Focus


Count Plot of transmission
16000
14000
12000
10000
8000
6000
4000
2000
0
count
Manual
Automatic
Semi-Auto
transmission


Count Plot of fuelType
count
12000
10000
8000
6000
4000
2000
0
Petrol
Diesel
Hybrid
Electric
Other
fuelType


year
price
mileage
tax
mpg
gineSize
1.00 0.64 -0.71 0.30 -0.02 -0.14
0.64 1.00 -0.53 0.41 -0.35 0.41
-0.71 -0.53 1.00 -0.26 0.12 0.21
0.30 0.41 -0.26 1.00 -0.50 0.19
-0.02 -0.35 0.12 -0.50 1.00 -0.26
-0.14 0.41 0.21 0.19 -0.26 1.00
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6

```
Target Feature:
price

Input Features:
['model', 'year', 'transmission', 'mileage', 'fuelType', 'tax', 'mpg', 'engineSize']

Before Encoding Shape: (17812, 9)
After Encoding Shape: (17812, 35)

First 5 Rows of Scaled Features:
       year   mileage       tax       mpg  engineSize  model_ C-MAX  \
0  0.067059 -0.382994  0.591380 -0.020597   -0.810561         False
1  0.554393 -0.736317  0.591380 -0.020597   -0.810561         False
2  0.067059 -0.562616  0.591380 -0.020597   -0.810561         False
3  1.041726 -0.665405  0.510777 -1.737858    0.345325         False
4  1.041726 -1.127749  0.510777 -0.908836   -0.810561         False

   model_ EcoSport  model_ Edge  model_ Escort  model_ Fiesta  ...  \
0            False        False          False           True  ...
1            False        False          False          False  ...
2            False        False          False          False  ...
3            False        False          False           True  ...
4            False        False          False           True  ...

   model_ Tourneo Connect  model_ Tourneo Custom  model_ Transit Tourneo  \
0                   False                  False                   False
1                   False                  False                   False
2                   False                  False                   False
3                   False                  False                   False
4                   False                  False                   False
```

```
   model_Focus  transmission_Manual  transmission_Semi-Auto  \
0        False                False                   False
1        False                 True                   False
2        False                 True                   False
3        False                 True                   False
4        False                False                   False

   fuelType_Electric  fuelType_Hybrid  fuelType_Other  fuelType_Petrol
0              False            False           False             True
1              False            False           False             True
2              False            False           False             True
3              False            False           False             True
4              False            False           False             True

[5 rows x 34 columns]

Final Feature Matrix Shape: (17812, 34)
Target Shape: (17812,)
```